Test Cases – Question Bank & Quiz Engine

This document covers all 89 test cases for the platform Question Bank & Quiz Engine

Summary

Section	Tests	Passed	Failed
1. User Authentication & Security	8	8	0
2. Profile Update & Approval Workflow	6	6	0
3. Audit Logging & Session Activity	4	4	0
4. Subject Syllabus & Question Quota Auto-Increase	6	6	0
5. Student Quiz Engine & Scorecards	4	4	0
6. Standardized Subject Codes & Auto-Versioning (New)	4	4	0
7. Subject Co-Teacher & Moderator Assignment (New)	3	3	0
8. Self-Moderation Prevention & Peer Auto-Assignment (New)	5	5	0
9. Security, Reliability & Load Testing	9	—	—
10. Req 6 – Auto-Grading	12	12	0
11. Req 7 – Analytics & Reporting	13	13	0
12. Req 8 – Security & Access Control	10	10	0
Total	**89**	**80***	**0**

1. User Authentication & Security

Test ID	Feature	Steps	Expected Result	Status
TC-01	BCrypt Password Hashing	Inspect PostgreSQL user table password_hash column.	Passwords begin with $2a$ / $2b$ BCrypt salt/hash. No plain text stored.	**Pass**
TC-02	Role-Based Dashboard Redirection	Log in as Admin (admin@mmcoe.edu.in), Author (kirtisatpute@mmcoe.edu.in), Moderator (geethasb@mmcoe.edu.in), Student (sahilveer...).	Redirects to correct role dashboard (admin.html, author.html, moderator.html, student.html).	**Pass**
TC-03	System Admin Credential Locking	Log in as Admin, then open My Profile.	Input fields are read-only (locked) with message "System Administrator credentials are locked."	**Pass**
TC-04	Invalid Credentials	Attempt login with wrong password.	Error message displayed: "Invalid email or password." Attempt logged in audit_log.	**Pass**
TC-05	First-Time Password Reset	Log in with initial account credentials.	Prompted to change password before proceeding to main dashboard.	**Pass**
TC-06	Session Expiry & Logout	Click Logout.	Session cleared, user redirected to login.html?logout=true.	**Pass**

Test ID	Feature	Steps	Expected Result	Status
TC-07	Multidevice LAN Access	Open http://<HOST_IP>:8080/student.html from a secondary laptop on the same Wi-Fi.	Dashboard loads cleanly and routes REST API calls to the host laptop.	**Pass**
TC-08	CORS & Security Headers	Send an OPTIONS preflight request to /api/auth/login.	Returns HTTP 204 with Access-Control-Allow-Origin: * headers.	**Pass**

2. Profile Update & Approval Workflow

Test ID	Feature	Steps	Expected Result	Status
TC-09	Profile Change Submission	Log in as Student/Teacher, update Division/Dept, then submit.	Request queued in localStorage.pendingProfileRequests and posted to /api/profile-requests.	**Pass**
TC-10	Admin Pending Approval Queue	Log in as Admin, then view Pending Approvals Modal.	Submitted profile update request appears with the requested changes.	**Pass**
TC-11	Immediate Approval Reflection	Admin clicks Approve.	User details (dept, division, contact, name) update instantly in the DB and local memory.	**Pass**
TC-12	Student Division & Dept Sync	Admin approves a Division change for a Student, then the student re-opens their profile.	Both division and department columns reflect the updated division value.	**Pass**
TC-13	Profile Request Rejection	Admin clicks Reject on a profile request.	Request status set to rejected; user profile remains unchanged.	**Pass**
TC-14	Admin Profile Lock Enforcement	Admin attempts to edit own credentials via the profile modal.	Save button disabled with "System Admin Locked."	**Pass**

3. Audit Logging & Session Activity

Test ID	Feature	Steps	Expected Result	Status
TC-15	Real-Time Audit Log Record	Log in as any user, then perform an action (e.g. approve QB, edit user).	Row inserted into PostgreSQL audit_log with user_id, action, and timestamp.	**Pass**
TC-16	Active Session Time Calculation	User logs in, stays active for 3 minutes, then logs out.	audit_log records the exact session duration (e.g., 3 mins or 45 secs).	**Pass**
TC-17	Audit Log CSV Export	Log in as Admin, go to the Audit Log tab, then click Export Audit Log (CSV).	CSV file downloads with complete activity history.	**Pass**
TC-18	Audit Log Filtering	Filter the Audit Log table by User or Action keyword.	Table filters dynamically without empty-state flickering.	**Pass**

4. Subject Syllabus & Question Quota Auto-Increase

Test ID	Feature	Steps	Expected Result	Status
TC-19	CSV Delimiter Auto-Detection	Upload a CSV using a comma, tab, or semicolon delimiter in create-question-bank.html.	PapaParse auto-detects the delimiter and parses 100% of rows without header corruption.	**Pass**
TC-20	DSML Question Quota Auto-Increase	Upload a 40-question DSML CSV as kirtisatpute@mmcoe.edu.in.	Chapter and Topic quotas increment in real time under Data Science and Machine Learning.	**Pass**

Test ID	Feature	Steps	Expected Result	Status
TC-21	1-to-1 Chapter Assignment	Upload a batch with various topics.	Each question is assigned to exactly one primary chapter (getBestChapterIndex()); 40 questions equal 40 total quota count.	**Pass**
TC-22	Live Topic Badges	Inspect the syllabus card on admin.html.	Each chapter displays dynamic topic badges with individual question counts (e.g. "Preprocessing – 13 Qs").	**Pass**
TC-23	Demo Quota Reset Button	Click "Reset Quotas to 0 (Demo)" on admin.html.	All subject question quotas zero out for clean team presentations.	**Pass**
TC-24	Descriptive Question Bank Live Visibility	Author publishes a Descriptive Question Bank, then a student logs in on student.html.	Descriptive question bank appears live on the student dashboard.	**Pass**

5. Student Quiz Engine & Scorecards

Test ID	Feature	Steps	Expected Result	Status
TC-25	Quiz Timer & Auto-Submit	Start quiz and allow the timer to count down to 0:00.	Quiz automatically submits the score and opens the results scorecard.	**Pass**
TC-26	Score & Grade Calculation	Complete a 10-question MCQ quiz (8 correct).	Scorecard displays 8/10 (80%), Grade A, and time taken.	**Pass**
TC-27	Instant Explanation View	View quiz results after submission.	Question explanations and reference textbook citations display under each question.	**Pass**
TC-28	Student Performance History	Student opens the Performance History modal.	Lists all completed attempts with dates, scores, and overall average.	**Pass**

6. Standardized Subject Codes & Auto-Versioning (New)

Test ID	Feature	Steps	Expected Result	Status
TC-29	Standard Code Format Validation (Valid)	Admin enters subject code CS301, IT402, or AIDS301 in the Add Subject modal.	Code accepted cleanly and auto-converted to uppercase.	**Pass**
TC-30	Standard Code Format Validation (Invalid)	Admin enters an invalid code: cs-301, 123CS, or C1.	Validation error displayed: "Subject code must be in standard format: 2-4 uppercase letters followed by 3 digits."	**Pass**
TC-31	Initial Syllabus Version Assignment	Create a new subject, CS305.	Syllabus version automatically assigned as v2026.1.	**Pass**
TC-32	Automatic Version Increment on Edit	Open CS301 (v2026.1), edit syllabus chapters/topics, then click Save.	Version automatically increments to v2026.2; manual typing not required.	**Pass**

7. Subject Co-Teacher & Moderator Assignment (New)

Test ID	Feature	Steps	Expected Result	Status
TC-33	Assign Co-Teachers During Subject Creation	Admin opens the subject modal and checks Ms. Kirti B. Satpute and Dr. Sankirti S. Shiravale.	Both faculty members saved as assigned co-teachers for the subject.	**Pass**

Test ID	Feature	Steps	Expected Result	Status
TC-34	Assign Designated Moderator	Admin selects Dr. Geetha R. Chillarge as Designated Moderator.	Moderator saved and displayed on the Subject Syllabus card.	**Pass**
TC-35	Syllabus Card Co-Teacher & Moderator Display	Inspect the Subject card on admin.html.	Displays badges: "Assigned Co-Teachers: Kirti Satpute, Sankirti Shiravale" and "Designated Moderator: Geetha Chillarge."	**Pass**

8. Self-Moderation Prevention & Peer Auto-Assignment (New)

Test ID	Feature	Steps	Expected Result	Status
TC-36	Co-Teacher Peer Auto-Assignment (Author A to Author B)	Log in as Ms. Kirti B. Satpute, then submit a DSML question bank.	System automatically sets assignedModerator to Dr. Sankirti S. Shiravale (the co-teacher).	**Pass**
TC-37	Co-Teacher Peer Auto-Assignment (Author B to Author A)	Log in as Dr. Sankirti S. Shiravale, then submit a DSML question bank.	System automatically sets assignedModerator to Ms. Kirti B. Satpute (the co-teacher).	**Pass**
TC-38	Self-Authored Table Row Badge	Log in as Kirti Satpute on moderator.html, then view the queue.	Kirti's own submitted question bank displays the badge "Self-Authored."	**Pass**
TC-39	Self-Moderation Action Button Disabling	Kirti Satpute opens the review modal for her own question bank.	Warning banner displayed: "Self-Moderation Restricted." Approve and Return buttons are disabled (disabled = true, opacity = 0.45).	**Pass**
TC-40	Peer Moderator Approval Execution	Log in as Sankirti Shiravale (or Geetha Chillarge) on moderator.html, then review Kirti's batch.	Approve button is fully enabled; batch approves successfully and status updates to approved.	**Pass**

9. Security, Reliability & Load Testing

Test ID	Feature	Steps	Expected Result	Status
TC-41	Security Headers Enforcement	Send an HTTP GET request to /api/subjects and inspect the response headers.	Response contains Strict-Transport-Security, X-Content-Type-Options: nosniff, and X-Frame-Options: DENY.	—
TC-42	Session Token Generation & Auth	Log in successfully, receive a session token, then call the protected API /api/admin/users.	Valid token allows access; a missing or invalid token returns HTTP 401 Unauthorized.	—
TC-43	Client-Server Heartbeat (apiClient.js)	Disconnect the server or block the network port, then re-enable the network.	apiClient.js detects offline status, displays a top-bar banner, and auto-reconnects when the server resumes.	—
TC-44	Login Rate Limiting (Attempts 1-4)	Enter an incorrect password for kirtisatpute@mmcoe.edu.in four consecutive times.	Warning message displays remaining allowed attempts (e.g. "1 attempt remaining before lockout").	—
TC-45	Login Lockout & HTTP 429 (Attempt 5)	Enter a fifth incorrect password for kirtisatpute@mmcoe.edu.in.	Account locked out for 15 minutes; API returns HTTP 429; login.html displays a real-time countdown timer.	—

Test ID	Feature	Steps	Expected Result	Status
TC-46	Lockout Expiry & Successful Login	Wait for the 15-minute lockout to expire (or trigger a demo reset), then log in with the correct password.	Lockout clears automatically; user logs in successfully to the dashboard.	—
TC-47	100 Simultaneous Active Quiz Takers	Execute simulate_100_quiz_students.py with 100 concurrent student threads.	Server processes all 100 quiz sessions simultaneously with 0 errors, a 100% submission rate, and average latency under 150 ms.	—
TC-48	Concurrent Author Question Bank Submissions	Execute simulate_concurrent_authors.py (Kirti Satpute and Sankirti Shiravale uploading for DSML at the exact same millisecond).	Both question banks saved cleanly; subject quotas update atomically; syllabus version auto-increments without race conditions or DB lock contention.	—
TC-49	Database Transaction Rollback on Failure	Simulate a network interruption midway through a 50-question bulk upload.	Database transaction rolls back completely (ROLLBACK), preventing partial or corrupted question bank entries.	—

10. Requirement 6 – Auto-Grading

Test ID	Test Title	Evidence	Status
TC6-01	Correct MCQ answer auto-grading	`HTTP 200 | Server Record: …`	**Pass**
TC6-02	Incorrect MCQ answer handling	`HTTP 200 | [{"id":"att_tc …`	**Pass**
TC6-03	True/False question auto-grading	`HTTP 200 | [{"id":"att_tc …`	**Pass**
TC6-04	Skipped question handling	`HTTP 200 | [{"id":"att_tc …`	**Pass**
TC6-05	Tampered client score protection	`HTTP 200 | Calculated pfS …`	**Pass**
TC6-06	Multiple answers per attempt database support	`HTTP 200 | Attempt att_tc …`	**Pass**
TC6-07	Quiz result persistence and retrieval	`HTTP 200 | Body: [{"id":" …`	**Pass**
TC6-08	Student access to personal attempt history	`HTTP 200 | Received JSON …`	**Pass**
TC6-09	Student data isolation and privacy protection	`HTTP 200 | Leaked Student …`	**Pass**
TC6-10	Subjective answer manual evaluation by faculty	`HTTP 200 | Body: [{"id":" …`	**Pass**
TC6-11	Unauthorized manual grading rejection	`HTTP 403 | {"status":"err …`	**Pass**
TC6-12	Accurate overall score and pass/fail calculation	`Passed standard: pctPass= …`	**Pass**

11. Requirement 7 – Analytics & Reporting

Test ID	Test Title	Evidence	Status
TC7-01	Dynamic assessment overview KPIs	`HTTP 200 | {"status":"suc …`	**Pass**
TC7-02	Graceful handling of empty analytics data	`HTTP 200 | Response lengt …`	**Pass**
TC7-03	Dynamic multi-cycle performance trends	`HTTP 200 | Output: [{"cyc …`	**Pass**

Test ID	Test Title	Evidence	Status
TC7-04	Trend metric computation accuracy	`HTTP 200 | Sample: [{"cyc …`	**Pass**
TC7-05	Psychometric item difficulty categorization	`HTTP 200 | [{"questionId" …`	**Pass**
TC7-06	Question discrimination index analysis	`HTTP 200 | [{"questionId" …`	**Pass**
TC7-07	Curriculum weak-topic identification	`HTTP 200 | [{"topic":"Gen …`	**Pass**
TC7-08	Multidimensional report dimension filtering	`HTTP 200 for all [lengths …`	**Pass**
TC7-09	Analytics report CSV format export	`HTTP 200 | Header: Class …`	**Pass**
TC7-10	Analytics report JSON format export	`HTTP 200 | Sample: [{"tit …`	**Pass**
TC7-11	Zero-dependency pure Java PDF report generation	`HTTP 200 | Content-Type: …`	**Pass**
TC7-12	Robustness with empty & unusual report filters	`HTTP 200 | Length: 1021 …`	**Pass**
TC7-13	Analytics data freshness upon new submission	`HTTP 200 | Fresh submissi …`	**Pass**

12. Requirement 8 – Security & Access Control

Test ID	Test Title	Evidence	Status
TC8-01	Unauthenticated analytics access rejection	`HTTP 401 | {"status":"err …`	**Pass**
TC8-02	Role-based authorization for administrative access	`Admin: HTTP 200 | Author: …`	**Pass**
TC8-03	Student privilege restrictions on admin endpoints	`analytics=403, audit=403, …`	**Pass**
TC8-04	Administrative full control validation	`users=200, audit=200`	**Pass**
TC8-05	Hardcoded demo token rejection in standard mode	`validateToken returned nu …`	**Pass**
TC8-06	BCrypt hashing and plaintext fallback elimination	`good=true, bad=false, pla …`	**Pass**
TC8-07	Login rate-limiting and account lockout	`HTTP 429 | {"status":"err …`	**Pass**
TC8-08	Session timeout and invalid token rejection	`HTTP 401 | {"status":"err …`	**Pass**
TC8-09	Question-bank answer key protection and authorization	`cPost=403, cDelete=403`	**Pass**
TC8-10	Security audit logging of critical actions	`HTTP 200 | Sample: [{"log …`	**Pass**